CS 301: Languages and Automata
Homework Assignment 02
Spring 2026

Instructions

This assignment is due <u>Sunday, February 22, at 11:59PM (Central Time)</u>.

This assignment must be submitted on *Gradescope*.

For Part I, you must submit JFLAP (.jff) files for problem 1. Each file must be <u>named properly</u> (1a.jff, 1b.jff, 1c.jff, etc.).

For Part II, you must submit answers for problems 2, 3, 4a, and 4b. <u>Answers must be typed</u> and submitted as a PDF file. Handwritten answers are <u>not</u> allowed. Additionally, each answer must be <u>clearly labeled and matched to the correct problem</u>. A <u>5-point penalty</u> will be applied to submissions that do not follow these guidelines.

Late submissions will be accepted within 0-12 hours after the deadline with a <u>5-point penalty</u> and within 12-24 hours after the deadline with a <u>10-point penalty</u>. No late submissions will be accepted more than 24 hours after the deadline.

<u>This assignment is individual</u>. Offering or receiving any kind of unauthorized or unacknowledged assistance (<u>including using generative AI or searching for solutions online</u>) is a violation of the University's academic integrity policies, will result in a grade of zero for the assignment, and will be subject to disciplinary action.

Part I: Regular Expressions (60 pt.)

1. **(60 pt., 10 pt. each) Construct a regular expression in JFLAP (<u>version 7.1</u>) that generates each of the following languages.**

   For each language, you must submit one **<u>properly named</u>** JFLAP (.jff) file: **1a.jff**, **1b.jff**, **1c.jff**, **1d.jff**, **1e.jff**, or **1f.jff**. **<u>Make sure that you test your regular expressions in JFLAP before submitting</u>**. A few strings that your regular expressions should and should not generate are given below as examples.

   Note that JFLAP uses the symbol **+** for union (instead of | or ∪) and the symbol **!** for the empty string (instead of ε). For more details about how to write regular expressions in JFLAP, see **JFLAP Tutorial 02** posted on Blackboard under **JFLAP Resources**.

   a. $\{\boldsymbol{w} \in \{\mathbf{a}, \mathbf{b}\}^* \mid \boldsymbol{w} \text{ does not end with } \mathbf{ab}\}$

   Your regular expression should be submitted as **1a.jff**.

   Generates: ba, aba, baa, aabb, baba

   Does not generate: ab, bab, aab, baab, ababab

b. $\{\boldsymbol{w} \in \mathbf{10^*1} \mid \text{the length of } \boldsymbol{w} \text{ is divisible by } 3\}$

Your regular expression should be submitted as **1b.jff**.

Generates: 101, 100001, 100000001, 100000000001, 100000000000001

Does not generate: 00, 1010, 01110, 010101, 0011100

c. $\{\boldsymbol{w} \in \{\mathbf{a}, \mathbf{b}\}^* \mid \boldsymbol{w} \text{ contains an odd number of } \mathbf{a}\text{'s or at most two } \mathbf{b}\text{'s}\}$

Your regular expression should be submitted as **1c.jff**.

Generates: bb, aaa, bababa, bbabb, bbbaaa

Does not generate: bbb, abbba, bbaabb, bbbaaaa, abababab

d. $\{\boldsymbol{w} \in \{\mathbf{0}, \mathbf{1}\}^* \mid \text{every odd position of } \boldsymbol{w} \text{ contains } \mathbf{0}\}$

Note: positions in a string are counted from left to right, starting at 1. For example, the string **0101** contains 0 in positions **1** and **3** and contains 1 in positions **2** and **4**.

Your regular expression should be submitted as **1d.jff**.

Generates: 01, 010, 0001, 01010, 010000

Does not generate: 10, 011, 0010, 10101, 000010

e. $\{\boldsymbol{w} \in \{\mathbf{a}, \mathbf{b}\}^* \mid \text{every } \mathbf{b} \text{ in } \boldsymbol{w} \text{ is both preceded and followed by } \mathbf{a}\}$

Your regular expression should be submitted as **1e.jff**.

Generates: aba, ababa, aabaa, abaaaba, aabaaaa

Does not generate: b, ab, baa, baab, abbaa

f. $\{\boldsymbol{w} \in \{\mathbf{0}, \mathbf{1}\}^* \mid \text{every substring } \mathbf{00} \text{ in } \boldsymbol{w} \text{ is followed by at least one } \mathbf{1}\}$

Your regular expression should be submitted as **1f.jff**.

Generates: 001, 0011, 11011, 001001, 1010010

Does not generate: 00, 100, 0001, 00100, 100011

Part II: Proofs (40 pt.)

2. (5 pt.) The following proof *incorrectly* concludes that $A = \{\mathbf{0}^m\mathbf{1}^n \mid m, n \geq \mathbf{0}\}$ is <u>not</u> a regular language, but A is regular. <u>Explain, in your own words, why this proof is incorrect</u>.

Proof (by contradiction). Assume that A is a regular language. Let p be the pumping length for A. Let s be the string 0^p1^p. By the pumping lemma, s can be divided into x, y, and z such that $|y| > 0$, $|xy| \leq p$, and $xy^iz \in A$ for any $i \geq 0$. Let $x = 0^p$, $y = 1$, and $z = 1^{p-1}$. Condition 2 of the pumping lemma does not hold because $|xy| = |0^p1| > p$. Therefore, the pumping lemma does not hold and A is not a regular language. *Q.E.D.*

3. (5 pt.) The following proof concludes that $B = \{\mathbf{0}^m\mathbf{1}^n \mid m \leq n \text{ and } m, n \geq \mathbf{0}\}$ is <u>not</u> a regular language, but does this *incorrectly*. <u>Explain, in your own words, why this proof is incorrect</u>.

Proof (by contradiction). Assume that B is a regular language. Let p be the pumping length for B. Let s be the string 0^p1^p. Consider all ways of dividing s into x, y, and z.

- **Case 1:** y contains only 0's. Then, $x = 0^m$, $y = 0^n$, and $z = 0^{p-m-n}1^p$ where $m \geq 0$ and $n > 0$. Condition 3 of the pumping lemma does not hold for this case because when $i = 0$, $xy^iz = xz = 0^{p-n}1^p \notin B$.
- **Case 2:** y does not contain only 0's. Then, $x = 0^{p-m}$, $y = 0^m1^n$, and $z = 1^{p-n}$ where $m \geq 0$ and $n > 0$. Condition 2 of the pumping lemma does not hold for this case because $|xy| = |0^p1^n| > p$.

Therefore, the pumping lemma does not hold for any case and B is not a regular language. *Q.E.D.*

4. **(30 pt., 15 pt. each) Prove that each of the following languages is <u>not</u> regular using the pumping lemma.**

   a. $\boldsymbol{C = \{0^m1^n \mid m \geq 2n \text{ and } m, n \geq 0\}, \Sigma = \{0, 1\}}$

   For example, $0000011 \in C$ and $00011 \notin C$.

   b. $\boldsymbol{D = \{x\#y \mid x, y \in \{\mathrm{a}, \mathrm{b}\}^* \text{ and } |x| = |y|\}, \Sigma = \{\mathrm{a}, \mathrm{b}\}}$

   For example, $\mathrm{aa\#bb} \in D$ and $\mathrm{aa\#b} \notin D$.